

Valid and efficient possibilistic inference in machine learning problems

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BFF8: Bayes, Fiducial, Frequentist

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- BFFs are largely model-centric:
 - likelihood (B and Fr)
 - data-generating equation (Fi , IMs, ...)
- Model-centricity is a bottleneck to BFF-style solutions to model-agnostic machine learning problems
- New perspectives needed for BFF-ML...

- *Why BFF-ML?* Uncertainty quantification!
- My claim:
 - reliable *probabilistic* UQ is too ambitious
 - but reliable *possibilistic* UQ is *possible*
- My goal here: sketch construction of a valid, model-agnostic *possibility distributions* for inference

- Problem setup
- Background
 - possibility theory basics
 - possibilistic inference with a likelihood²
- Reliable, model-agnostic, possibilistic machine learning
 - prediction of future observations
 - inference on risk minimizers (teaser)
- Concluding remarks

²I'm sure you don't remember, but I talked about this at BFF7...

- Data $Z^n \in \mathbb{Z}^n$, observed value z^n
- ML applications: often z is a feature–label pair (x, y)
- Joint probability distribution P for Z^n
- Uncertain quantity of interest is $\Theta \in \mathbb{T}$:
 - a new observation $\Theta = Z_{n+1}$
 - a functional $\Theta = \phi(P)$, e.g., risk minimizer
- **Goal:** *quantify uncertainty about Θ , given $Z^n = z^n$*
- That is, I want to construct a reliable, probability/possibility distribution for Θ , given observed $Z^n = z^n$

- *Why possibility theory?* I'll explain later...
- Possibility distribution for generic uncertain T in $\mathbb{T}$
 - determined by a *possibility contour* $\pi : \mathbb{T} \rightarrow [0, 1]$
 - technical requirement: $\sup_{t \in \mathbb{T}} \pi(t) = 1$
- Possibility measure,³ aka consonant plausibility function

$$\bar{\Pi}(A) = \sup_{t \in A} \pi(t), \quad A \subseteq \mathbb{T}$$

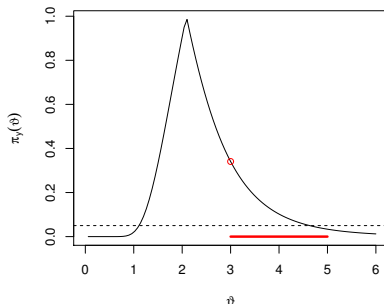
- Credal set $\mathcal{C}(\bar{\Pi}) = \{Q \in \text{probs}(\mathbb{T}) : Q(\cdot) \leq \bar{\Pi}(\cdot)\}$
- $\mathcal{C}(\bar{\Pi})$ is really important,⁴ but no time to explain :(

³Corresponding necessity measure $\underline{\Pi}(A) = 1 - \bar{\Pi}(A^c)$

⁴e.g., $\bar{\Pi} \iff \mathcal{C}(\bar{\Pi})$ which aids interpretation

Possibility theory, cont.

- Possibility contour π
- $A = [3, 5]$
- $\bar{\Pi}(A)$ via maximization
 - most possible member
 - size of A is irrelevant*
- $\{\theta : \pi(\theta) > 0.05\}$



* $\bar{\Pi}$ is a differential-element-free “measure” (M. [arXiv:2303.17425](https://arxiv.org/abs/2303.17425))

- Not all credal sets correspond to possibility measures
- Given a credal set $\mathcal{Q}$, find a best outer consonant approx
 - a possibility measure $\bar{\Pi}$ with $\mathcal{C}(\bar{\Pi}) \supseteq \mathcal{Q}$
 - and $\mathcal{C}(\bar{\Pi})$ is “as small as possible”
- Given a *ranking function* $r : \mathbb{T} \rightarrow \mathbb{R}$, there is one best outer consonant approximation compatible with r
- Solution given by the *probability-to-possibility transform*⁵

$$\pi(t) = \sup_{Q \in \mathcal{Q}} Q\{r(T) \leq r(t)\}, \quad t \in \mathbb{T}$$

- This is the workhorse for a new brand of IMs

⁵Dominik Hose's PhD thesis, 2022; papers by Dubois & Prade

Possibilistic inferential models (IMs)

- Classical statistical inference setup:
 - a model $\{P_\theta : \theta \in \mathbb{T}\}$ for Z^n given $\Theta = \theta$
 - a class $\mathcal{Q}$ of priors Q for Θ
- No obvious probabilistic UQ
- Possibility distribution for Θ , given $Z^n = z^n$?
- Idea: treat (Z^n, Θ) as the generic T above
- But “given $Z^n = z^n$ ” requires some new maneuvers...

Valid and efficient imprecise-probabilistic inference
with partial priors, II. General framework

Ryan Martin*

arXiv:2211.14567

- Model determines an upper joint dist'n⁶ for (Z^n, Θ) :

$$\bar{P}_{\mathcal{Q}}(Z^n \in B, \Theta \in A) = \sup_{Q \in \mathcal{Q}} \int_A P_{\theta}(Z^n \in B) Q(d\theta)$$

- $r(z^n, \theta) =$ “normalized upper density for (Z^n, Θ) ”
- IM's possibility contour (depending on r , $\mathcal{Q}$, etc)

$$\pi_{z^n}(\theta) = \bar{P}_{\mathcal{Q}}\{r(Z^n, \Theta) \leq r(z^n, \theta)\}, \quad \theta \in \mathbb{T}$$

- Induced possibility distribution for Θ , given $Z^n = z^n$, i.e.,

$$\bar{\Pi}_{z^n}(A) = \sup_{\theta \in A} \pi_{z^n}(\theta), \quad A \subseteq \mathbb{T}$$

- There's a corresponding credal set $\mathcal{C}(\bar{\Pi}_{z^n})\dots$

⁶This is a Choquet integral

Theorem.

Possibilistic IM is *valid* in the sense that $\bar{P}_{\mathcal{Q}}\{\pi_{Z^n}(\Theta) \leq \alpha\} \leq \alpha$ for all $\alpha \in [0, 1]$. Consequently,

- (a) if $C_\alpha(z^n) = \{\theta : \pi_{Z^n}(\theta) > \alpha\}$, then $\bar{P}_{\mathcal{Q}}\{C_\alpha(Z^n) \not\subseteq \Theta\} \leq \alpha$
- (b) $\bar{P}_{\mathcal{Q}}(\bigcup_{A:A \ni \Theta} \{\bar{\Pi}_{Z^n}(A) \leq \alpha\}) \leq \alpha$

- *Why possibility theory?* These results aren't possible w/o it
- Consequences:
 - (a) elements of $\mathcal{C}(\bar{\Pi}_{Z^n})$ are confidence distributions⁷
 - (b) *uniformity in A* is related to the IM's probativeness⁸
- Also some coherence/no-sure-loss properties⁹

⁷M., arXiv:2303.08630

⁸Cella and M., arXiv:2304.05740

⁹M., arXiv:2211.14567

- Often there's simplifying structure in the problem
- *Principle of Minimal Complexity*:¹⁰ Reduce dimension!
- Applications of the principle involve *conditioning*
- Example:
 - frequentists: $\mathcal{Q} = \{\text{all priors}\}$
 - $r(z^n, \theta) = \text{relative likelihood}$
 - problem allows for *conditioning* on $\Theta = \theta$:

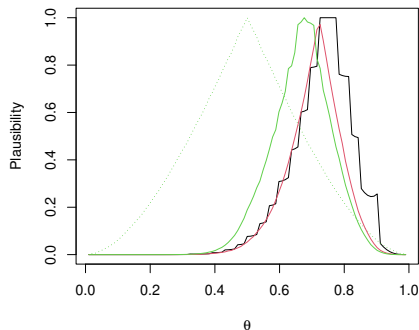
$$\pi_{z^n}(\theta) = P_{\theta}\{r(Z^n, \theta) \leq r(z^n, \theta)\}, \quad \theta \in \mathbb{T}$$

- this is familiar... it's a p-value for $H_0 : \Theta = \theta$
- same validity properties, but more efficient

¹⁰M., arXiv:2211.14567

Possibilistic IMs, cont.

- Binomial model: $P_\theta = \text{Bin}(n, \theta)$, for fixed $n = 16$
- IM possibility contour (based on 12 successes)
 - frequentist “every prior”
 - Bayesian “one prior”
 - IM “partial prior”



- ML applications don't have models
- Above construction depends on model in
 - the ranking function
 - the “prob” in my prob-to-poss transform
- But maybe we don't actually need those things...
- Let's focus on a prediction setting
 - Z^n is observable, interest in $\Theta = Z_{n+1}$
 - assume exchangeable with unknown dist'n P
 - set of probabilities, $\mathcal{P} = \{\text{exchangeable } P\}$
- $Z = (X, Y)$ case is similar, just messier notation

- Let $\gamma : \mathbb{Z}^n \times \mathbb{Z} \rightarrow \mathbb{R}$ be a measure of *conformity*
 - large values of $\gamma(z^n, z_{n+1})$ mean z_{n+1} “conforms with” z^n
 - e.g., $\gamma(z^n, z_{n+1}) = -d(\hat{z}_{z^n}, z_{n+1})$
 - assume invariant to permutations of its first argument
- Let's use γ as our ranking function
- Naive application of the prob-to-poss transform suggests

$$\pi_{z^n}(z_{n+1}) = \sup_{P \in \mathcal{P}} P\{\gamma(Z^n, Z_{n+1}) \leq \gamma(z^n, z_{n+1})\}$$

- Corresponding IM has all the nice properties, but...
- A practical solution requires a complexity reduction

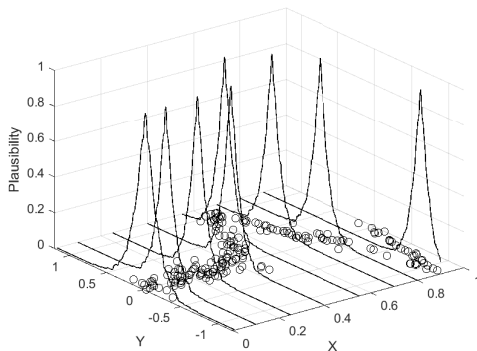
- *Principle of Minimal Complexity*... conditioning
- Condition on the set of values¹¹ $\{z_1, \dots, z_n, z_{n+1}\}$

$$\pi_{z^n}(z_{n+1}) = \sup P[\gamma(Z^n, Z_{n+1}) \leq \gamma(z^n, z_{n+1}) \mid \{z^n, z_{n+1}\}]$$
$$\text{sufficiency} \rightarrow = \frac{1}{(n+1)!} \sum_{\sigma} 1\{\gamma(z^{\sigma(1:n)}, z_{\sigma(n+1)}) \leq \gamma(z^n, z_{n+1})\}$$
$$\text{invariance} \rightarrow = \frac{1}{n+1} \sum_{i=1}^{n+1} 1\{\gamma(z_{-i}^{n+1}, z_i) \leq \gamma(z^n, z_{n+1})\}$$

- RHS is Vovk & Shafer's conformal prediction “transducer”

¹¹Faulkenberry (*JASA* 1973), Hoff (*Bernoulli* 2023)

- For feature-label pairs $Z = (X, Y)$, IM produces possibility contour functions for Y at each $X = x$



References (from before I knew about prob-to-poss)

 International Journal of Approximate Reasoning 
Volume 141, February 2022, Pages 110-130

Validity, consonant plausibility measures, and conformal prediction

[Leonardo Cella](#)  , [Ryan Martin](#) 

 International Journal of Approximate Reasoning 
Volume 150, November 2022, Pages 1-18

Valid inferential models for prediction in supervised learning problems ☆

[Leonardo Cella](#)  , [Ryan Martin](#) 

- For model-agnostic inference on risk minimizers, can use the empirical risk for ranking
- *Bootstrap* provides a low-complexity approximation
- Reference (before I knew prob-to-poss)



Conclusion

- Possibilistic IMs can do everything (?) we want
- Probability-to-possibility transform is the workhorse
- Here I showed that conformal prediction is a consequence
- More BFF–ML developments are *possible*
- Want to learn more about imprecise prob in stat?



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

[https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/](https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/)

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